Assignment

Submission Instructions:

- The work must be written on an official sheet.
- Must be submitted on Wednesday, 12 August 2026.

1. Figure 1 is a transshipment network with four source nodes (1–4), three intermediary nodes (5–7), and three destination nodes (8–10). The supplies are 1000, 800, 1150, and 2050 units respectively, and the demands are 1300, 2000, and 1700 units respectively. The unit transportation costs are displayed along each arc in the network.

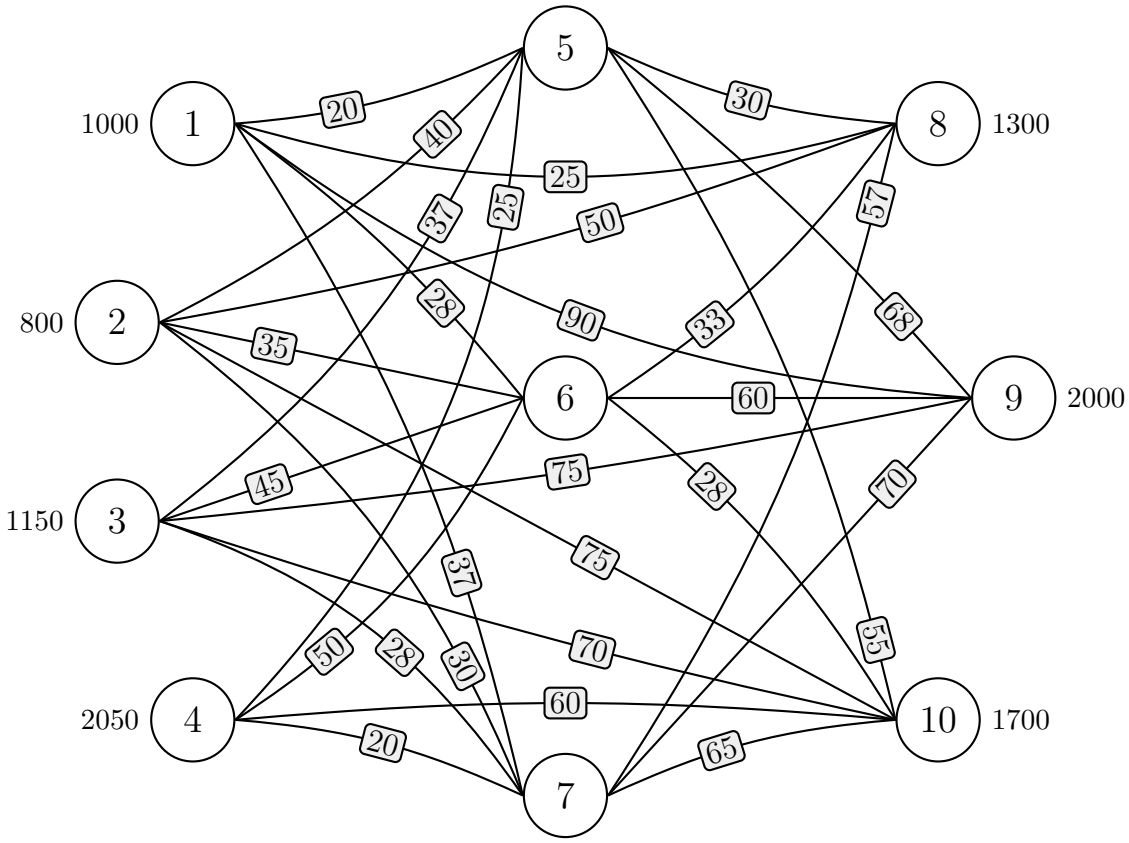


Figure 1: The transport network.

Based on the transshipment network presented above, complete the following tasks:

(a) Formulate the complete mathematical model for this transshipment network.

(b) Construct the standard transportation tableau corresponding to this transshipment problem.

(c) Use ADBAM to find the initial basic feasible solution .

(d) Use the uv method to find the optimal solution.

2. A farmer has 1,000 acres of land on which he can grow corn, wheat or soyabean. Each acre of corn costs Rs 100 for preparation, requires 7 men-days of work and yields a profit of Rs 30. An acre of wheat costs $120 to prepare, requires 10 men-days of work and yields a profit of $ 40. An acre of soyabean costs $70 to prepare, requires 8 men-days of work and yields a profit of $20. If the farmer has $100,000 for preparation and can count on 8,000 men-days of work, determine how many acres should be allocated to each crop in order to maximize profits?

3. A company produces three products A, B and C. These products require three ores O_1, O_2 and O_3. The maximum quantities of the ores O_1, O_2 and O_3 available are 22 tonnes, 14 tonnes and 14 tonnes, respectively. For one tonne of each of these products, the ore requirements are:

	A	B	C
O_1	3	–	3
O_2	1	2	3
O_3	3	2	3
Profit per tonne	1000	4000	5000

   (a) How many tonnes of each product should the company produce in order to maximize its profits.

   (b) Determine the range of feasibility for the right-hand side value of ore O_2.

   (c) Determine the range of optimality for the profit coefficient of product A.